5 A capacitor of capacitance C is connected in parallel with a resistor of resistance 200 Ω across an alternating supply as shown in Fig. 5.1.

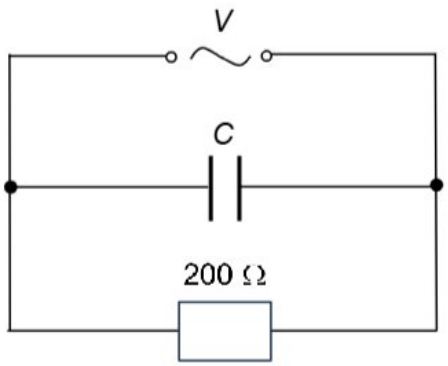


Fig. 5.1

The variation with time t of the potential difference V of the alternating supply is given by

$$V = 120\sin(314\,t),$$

where V is measured in volts and t in seconds.

(a) (i) Write an equation, in terms of C, that relates how charge Q on the capacitor plate varies with t.

[1]

(ii) Hence, show the current I_C through the capacitor is $37700C\cos(314\,t)$.

[1]

(b) (i) Determine the r.m.s. current $I_{R,rms}$ through the resistor.

$I_{R,rms}$ = ... A [1]

(ii) Calculate the mean power P dissipated in the resistor.

P = ... W [2]

(c) The frequency of V is increased.

State and explain the effect, if any, on

(i) the r.m.s. current through the resistor,

.....

.....

..... [1]

(ii) the r.m.s. current through the capacitor.

.....

.....

..... [1]

[Total: 7]